

Syllabus

Spring semester 2026

Course Title	Differential Equations		Course Code		EMC3113		Credit	3	Classroom	H3511
			Type		Advanced Elective		Hours	4		
Teacher	Name	Class no.	Office	Office Hour*		Contact		E-mail		
	Hyojin Kim	1	H3502	Mon 2,6,7		051-606-2385		hyojinkim@ksa.kaist.ac.kr		

*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Descriptions

Course Description

This course introduces the theory and techniques of ordinary and partial differential equations. Topics covered include first-order differential equations, higher-order linear differential equations, series solutions, Laplace transforms, systems of differential equations, and an introduction to partial differential equations.

Course Objectives

- Understand the fundamental concepts and techniques of solving ordinary differential equations.
- Gain proficiency in solving various types of differential equations using analytical and numerical methods.
- Develop an understanding of the applications of differential equations in various fields such as physics, engineering, and biology.
- Be able to analyze and interpret solutions to differential equations in practical contexts.

2. Core Target Competencies

Subject Core Competencies

Conceptual Understanding	Computational Accuracy	Communication	Reasoning & Thinking Skills	Application
Conceptual Understanding: The ability to accurately understand mathematical definitions, concepts, and properties, and to identify relationships with new concepts based on this understanding.				
Computational Accuracy: The ability to solve problems or compute values accurately and efficiently using mathematical procedures and algorithms.				
Communication: The ability to clearly express mathematical ideas, solution processes, and problem-solving strategies through speech, writing, or visual representations, and to effectively communicate with others.				
Reasoning & Thinking: The ability to logically analyze and evaluate mathematical ideas and solution strategies, make valid inferences, and construct sound arguments in problem-solving contexts.				
Application: The ability to apply mathematical ideas and strategies to new or real-world situations by transferring knowledge and integrating different concepts effectively.				

3. Textbooks and References

Elementary Differential Equations and Boundary Value Problems, (12th edition, International Adaptation)
WILLIAM E. BOYCE & RICHARD C. DIPRIMA & DOUGLAS B. MEADE, WILEY & Sonc Inc

4. Class Method

- A. Explanation of course content by the instructor
- B. Student presentations and participation in class
- C. Assessment through assignments, quizzes, and exams

5. Evaluation

A. Evaluation Criteria

Midterm exam	Final exam	Attendance	Quizzes	Participation & Homework	Sum
30%	40%	10%	10%	10%	100%

B. Evaluation Method

(1) Midterm Exam, Final Exam

- Written exams will be administered during the semester. Covered sections will be informed later.

(2) Attendance

- A penalty of 1 point will be given for tardiness and 3 points for absence.

(3) Quizzes

- There will be two in-class quizzes in this semester.

(4) Other Activities

- Assignments or presentations in class
- Subject to change depending on the instructors.

6. Criteria for Awarding Credits

The A-F grades will be given based on overall students performance.

Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal’s Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

서식 있음: 눈금에 맞춤, 탭: 30.08 글자, 왼쪽 + 8 글자(없음) + 12 글자 + 16 글자 + 20 글자 + 24 글자 + 28 글자 + 32 글자 + 36 글자 + 40 글자 + 44 글자 + 48 글자 + 52 글자 + 56 글자 + 60 글자 + 64 글자 + 68 글자 + 72 글자 + 76 글자 + 80 글자 + 84 글자 + 88 글자 + 92 글자 + 96 글자 + 100 글자 + 104 글자 + 108 글자 + 112 글자 + 116 글자 + 120 글자 + 124 글자 + 128 글자

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- Calculation of the basis score for awarding credit to absentees : In case of an examination missed due to official leave or medical leave, the score will be calculated by multiplying (the score corresponding to the total percentile score earned by the student on other examinations in the same subject) by (100% for official leave, or 80% for medical leave).
- If a basis score cannot be calculated, the grading method : If no reference score is available, a re-evaluation will be conducted.
- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	1.1-1.3 Introduction 2.1 Linear Differential Equations 2.2 Separable Differential Equations 2.6 Exact Differential Equations and Integrating Factors	
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 2 3.2.~3.8.	2.4 Difference Between Linear and Nonlinear Differential Equations, Bernoulli Equations, Ricatti Equations 2.5 Autonomous Differential Equations and Population Dynamics 2.8 The Existence and Uniqueness Theorem	3.2. Substitute Holiday
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 3 3.9.~3.15.	3.1 Homogeneous Differential Equations with Constant Coefficients 3.2 Solutions of Linear Homogeneous Equations; the Wronskian 3.3 Complex Roots of the Characteristic Equations 3.4 Repeated Roots: Reduction of Order	
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 4 3.16.~3.22.	3.5 Nonhomogeneous Equations; Method of Undetermined Coefficients 3.6 Variation of Parameters	

	4.1 General Theory of nth Order Linear Differential Equations 4.2 Homogeneous Differential Equations with Constant Coefficients	
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 5 3.23.~3.29.	4.3 The Method of Undetermined Coefficients 4.4 The Method of Variation of Parameter 5.1 Review of Power Series	Quiz 1
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 6 3.30.~4.5.	5.2 Series Solutions First Order Equations 5.3 Series Solutions Near an Ordinary Point I 5.4 Series Solutions Near an Ordinary Point II	
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 7 4.6.~4.12.	5.5 Euler Equations: Regular Singular Points 5.6 Series Solutions Near a Regular Singular Point I 5.7 Series Solutions Near a Regular Singular Point II	
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	7.1 Definition of the Laplace Transform 7.2 Solution of Initial Value Problems 7.3 Step Functions 7.4 Differential Equations with Discontinuous Forcing Functions	
Week 10 4.27.~5.3.	7.5 Impulse Functions 7.6 The Convolution Integral 6.1 Introduction	
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 11 5.4.~5.10.	6.2 Matrices 6.3 System of Linear Algebraic Equations 6.4 Basic Theory of Systems of First-Order Linear Equations	5.5. Children's Day
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	

	Communication, Reasoning & Thinking, Application	
Week 12 5.11.~5.17.	6.5 Homogeneous Linear Systems with Constant Coefficients 6.6 Complex-Valued Eigenvalues 6.7 Fundamental Matrices	
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 13 5.18.~5.24.	6.8 Repeated Eigenvalues 6.9 Nonhomogeneous Linear Systems 9.1 The Phase Plane: Linear Systems	Quiz 2
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 14 5.25.~5.31.	9.2 Autonomous Systems and Stability 9.3 Locally Linear Systems 9.4 Competing Species	5.25. Substitute Holiday
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 15 6.1.~6.7.	9.5 Predator-Prey Equations 9.6 Liapunov's Second Method 9.7 Periodic Solutions and Limit Cycles	6.3. Local Election Day
Core Competencies	♦ Conceptual Understanding, Computational Accuracy, Communication, Reasoning & Thinking, Application	
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/Quantitative Literacy	5. Global Perspective	6. Adaptability/Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect creativity	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasticity
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/patience
Ability to devise and test solutions to problems by	Ability to explain information and persuasively	Ability to collaborate on tasks, manage groups, and		Ability to use 21 st century core competencies such as communication	Ability to develop a sense of independence to seek new roles,	Ability to make rational and ethical decisions on complex	Self-efficacy

서식 있음: 왼쪽, 단어 잘림 허용

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/Quantitative Literacy	5. Global Perspective	6. Adaptability/Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/Mindset
trial and error	engage others in an engaging way	delegate responsibilities		skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	ideas, or strategies	issues	
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex real-world problems							

서식 있음: 왼쪽, 단어 잘림 허용

1. Complex Problem-Solving Problem-Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, <u>and</u> systematic and efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and Decision-making Skills	Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways